

MATH 345 Skeleton Notes

Unit 2: Anatomy of a linear map

Section 2.1: Understanding a linear map: a preview

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In Vectors, matrices, and linear maps, a matrix defined a linear map

$$\mathbf{x} \mapsto A\mathbf{x}.$$

This unit answers the following questions about the behavior of this map. The second column previews the concepts introduced later in the unit for this purpose.

- **Question:** When do different inputs give the same output?
Concept/Method: $A(\mathbf{x} + \mathbf{z}) = A\mathbf{x}$
- **Question:** What input changes does A forget?
Concept/Method: $\text{null}(A)$
- **Question:** When can a desired output be hit?
Concept/Method: $A\mathbf{x} = \mathbf{b}$
- **Question:** What outputs can A produce?
Concept/Method: $\text{im}(A)$
- **Question:** Can the map be reversed?
Concept/Method: A^{-1} , when it exists

We give several preliminary examples.

Activity: A map that forgets height

Let

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

First compute $P\mathbf{x}_1$, $P\mathbf{x}_2$, and $P\mathbf{x}_3$, where

$$\mathbf{x}_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix}.$$

1. Which input coordinate changed?
2. Which part of the input did the output remember?
3. Which part of the input did the output forget?
4. Suppose the desired output is $(2, 1)$. What simple equations must the input coordinates satisfy?
5. Following up on the previous question, how many choices of the third coordinate work?

Activity: The same height loss in code

The following code repeats the same experiment.

```
import numpy as np

P = np.array([[1, 0, 0],
              [0, 1, 0]])

x1 = np.array([4, -2, 0])
x2 = np.array([4, -2, 6])
x3 = np.array([4, -2, -5])

P @ x1, P @ x2, P @ x3
```

Predict the three code outputs.

Activity: Can this map produce the requested output?

Let

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}.$$

First compute $A\mathbf{x}_1$, $A\mathbf{x}_2$, and $A\mathbf{x}_3$, where

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

1. Can this map produce the output $(1, 2, 3)$? If yes, give an input.
2. What rule do the outputs above all obey?
3. For the desired output $(1, 2, 4)$, what equations would the unknown input $\mathbf{x} = (x, y)$ have to satisfy?
4. Following up on the previous question, can all of those equations be true at the same time?

Activity: Two inputs, one output

Let

$$F = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

First compute $F\mathbf{u}$ and $F\mathbf{v}$, where

$$\mathbf{u} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix}.$$

1. Are the inputs the same?
2. Are the outputs the same?
3. Specify $\mathbf{d}$ such that $\mathbf{v} = \mathbf{u} + \mathbf{d}$.
4. What equations does $\mathbf{d} = (d_1, d_2, d_3)$ satisfy if $F\mathbf{d} = \mathbf{0}$?

Activity: Same output in code

The following code compares two more inputs for the same map.

```
F = np.array([[1, 1, 0],  
              [0, 1, 1]])  
  
u = np.array([0, 1, 2])  
v = np.array([1, 0, 3])  
d = v - u  
  
F @ u, F @ v, d, F @ d
```

Predict the four code outputs.

Activity: Output scores and a hidden direction

A final model layer can convert a hidden vector into output scores by

$$\ell = W\mathbf{h} + \mathbf{b}.$$

Let

$$W = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 0.5 \\ -0.5 \end{bmatrix}, \quad \mathbf{h} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{d} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}.$$

1. Compute $W\mathbf{h} + \mathbf{b}$.
2. Compute $W\mathbf{d}$.
3. Use part 2 to compute $W(\mathbf{h} + \mathbf{d}) + \mathbf{b}$.
4. Are $\mathbf{h}$ and $\mathbf{h} + \mathbf{d}$ the same hidden vector?
5. Are their output-score vectors the same?
6. Did the bias vector $\mathbf{b}$ affect which hidden-vector change was invisible?

Here is the same issue in a data-matrix setting.

Activity: Two coefficient choices, same predictions

A data matrix stores three measured features for four observations:

$$X = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 1 \\ 1 & -1 & 0 \end{bmatrix}.$$

Observe that the third feature is the sum of the first two features in every row.

Consider the two coefficient vectors

$$\mathbf{c} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{c}' = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

In this example, the linear map takes a coefficient vector $\mathbf{c}$ as input and returns the prediction vector $X\mathbf{c}$.

1. Compute $X\mathbf{c}$.
2. Compute $X\mathbf{c}'$.
3. Are the coefficient vectors the same?
4. Are the predictions the same?
5. Let $\mathbf{d} = \mathbf{c} - \mathbf{c}'$. What equations does $X\mathbf{d} = \mathbf{0}$ represent?

Each preview led to the same kind of question: can certain linear equations in the unknown input coordinates be solved? The next section

gives this question a formal name and a systematic method.